Colour Centers

Optical Spectroscopy of Colour Centers and Molecules
(Frank-Condon Principle)

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1 Introduction

In this experiment, we measured transmission spectra.

1.1 Molecular Spectra

1.1.1 Electronic Transitions

Electronic transitions occur when a molecule absorbs or emits a photon, leading to a change in its electronic energy state. The energy of the absorbed or emitted photon is given by:

$$E = \frac{hc}{\lambda} \tag{1}$$

In diatomic molecules, solving the Schrödinger equation is more complex than for atoms due to the Coulomb interaction between the nuclei, which breaks spherical symmetry and introduces axial symmetry. This symmetry reduction lifts the degeneracy of energy levels and results in distinct electronic, vibrational, and rotational states, allowing for a rich spectrum of transitions.

1.1.2 Vibrational Transitions

In order to model the vibrational transitions we consider the diatomic molecule as a harmonic oscillator as a first approximation.

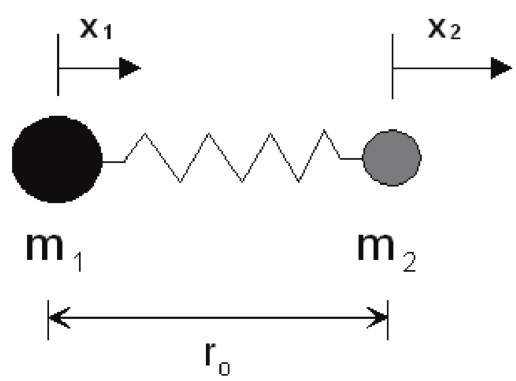


Figure 1: Harmonic Oscillator model [2]

Equations of motion become:

$$m_1\ddot{x}_1 = -k(x_1 - x_2)$$
$$m_2\ddot{x}_2 = -k(x_2 - x_1)$$

Since the part of the system we are interested in has 1 degree of freedom, we do with $x = x_1 - x_2$ and also $\mu = \frac{m_1 m_2}{m_1 + m_2}$ to get

$$\ddot{x} = -\frac{k}{\mu}x$$

For a harmonic oscillator, the fundamental vibrational wavenumber is given by:

$$\bar{\nu}_s = \frac{1}{2\pi c}\sqrt{\frac{k}{\mu}}$$

where:

- k is the force constant of the bond,
- μ is the reduced mass of the diatomic system,
- c is the speed of light.

To find the energy levels, we solve the Schrödinger equation using the harmonic potential:

$$U(x) = \frac{1}{2}kx^2 = 2\pi^2 c^2 \bar{\nu}^2 \mu x^2$$

This potential arises from the restoring force:

$$F = -kx = -Ux$$

Assuming a classical solution of the form:

$$x(t) = x_0 \sin(2\pi c \bar{\nu}_s t)$$

the quantized energy levels are:

$$E_n = hc\bar{\nu}_s \left(n + \frac{1}{2}\right), \quad n = 0, 1, 2, \ldots$$

and transitions follow the selection rule:

$$\Delta n = \pm 1$$

However, experimental observations show that the vibrational level spacing decreases as n increases. Moreover, at large bond lengths, molecular dissociation occurs, and at very short distances, nuclear repulsion becomes significant. These limitations make the harmonic oscillator model less accurate at high energies.

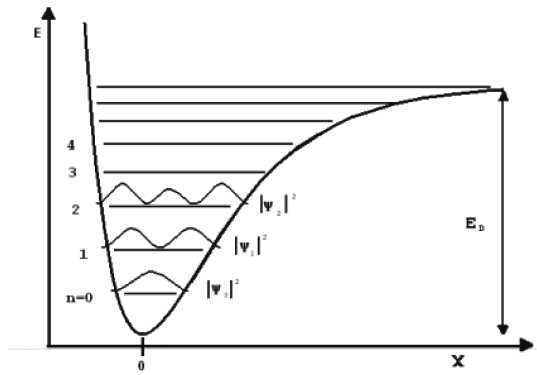


Figure 2: Anharmonic oscillator with vibrational energy levels [2]

To account for these effects, the Morse potential provides a better description:

$$U(x) = E_D a^2 x^2 \quad \text{(for small } x\text{)}$$

where:

- E_D is the dissociation energy,
- a is a molecular-specific constant.

The parameter a is given by:

$$a = 2\pi\bar{\nu}_s c\sqrt{\frac{\mu}{2E_D}} = \omega_s\sqrt{\frac{\mu}{E_D}}$$

Solving the Schrödinger equation with the Morse potential leads to energy levels approximated by:

$$E_n = hc\bar{\nu}\left[\left(n+\frac{1}{2}\right) - \gamma_e\left(n+\frac{1}{2}\right)^2\right]$$

where γ_e is the anharmonicity constant. We get the following diagram for energy levels of an anharmonic oscillator.

1.1.3 Combined Spectra

Now were consider the case when both electronic and vibrational transitions apply.

Going from a the ground electronic state to excited one there are 3 possibilities. The excited state could have the same binding energy (a), lower binding energy (b), or higher binding energy (c).

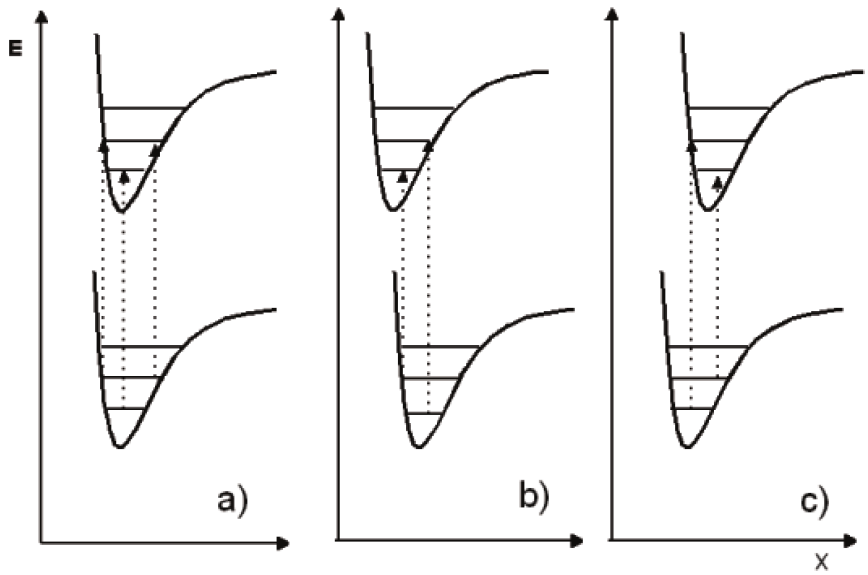


Figure 3: Possible Potential curves [2]

For vibrational transitions between levels n'' in the lower electronic state and n' in the upper electronic state, the transition energy is given by:

$$\begin{aligned}\Delta E(n', n'') &= E(n') - E(n'') \\ &= E_{\text{elektr}} + hc\bar{\nu}_s' \left(n' + \frac{1}{2}\right) - hc\bar{\nu}_s'' \left(n'' + \frac{1}{2}\right) \\ &\quad - hc\bar{\nu}_s'\gamma_e' \left(n' + \frac{1}{2}\right)^2 + hc\bar{\nu}_s''\gamma_e'' \left(n'' + \frac{1}{2}\right)^2 \end{aligned} \tag{2}$$

Although the dissociation energy E_D represents the energy required to break the bond completely, a direct transition to this energy is unlikely in practice. This is because transitions involving large differences between n' and n'' (i.e., large Δn) are highly improbable due to selection rules.

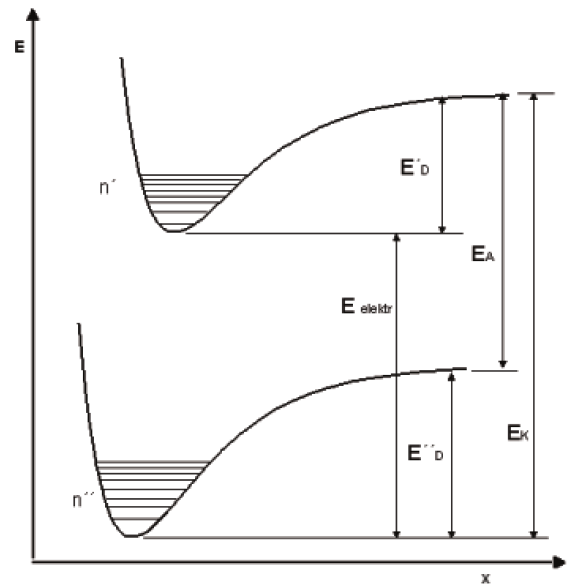


Figure 4: Definition of Transition Energies [2]

$$E = \frac{hc}{\lambda} \tag{3}$$

When an electron is excited to a higher electronic state, the probability of vibrational transitions increases. The energy difference between vibrational levels n' and n'' in different electronic states is given by:

$$\Delta E = E_{\text{electronic}} + E'_S \left(n' + \frac{1}{2}\right) - E'_S \gamma'_e \left(n' + \frac{1}{2}\right)^2 \tag{4}$$

For large values of n', the transition energy ΔE approaches the dissociation energy E_K. The difference between two vibrational transitions with a quantum number difference N is:

$$\begin{aligned}\Delta(\Delta E) &= E'_S \left(n' + \frac{1}{2} + N\right) - E'_S \left(n' + \frac{1}{2}\right) \\ &\quad - E'_S \gamma'_e \left(n' + \frac{1}{2} + N\right)^2 + E'_S \gamma'_e \left(n' + \frac{1}{2}\right)^2 \\ &= E'_S N - E'_S \gamma'_e N(2n' + N + 1)\end{aligned} \tag{5}$$

For $N = 1$, this simplifies to:

$$\Delta(\Delta E) = E'_S \left[1 - \gamma'_e (2n' + 2)\right] \tag{6}$$

This expression shows that the spacing between neighboring vibrational transitions decreases linearly with increasing n'. The difference in spacing between two successive transitions is constant:

$$\Delta(\Delta E)(n') - \Delta(\Delta E)(n'+1) = -2E'_S\gamma'_e = \text{const.} \tag{7}$$

As this difference approaches zero near the dissociation energy E_K, the vibrational quantum number at dissociation can be estimated as:

$$n'_E = \frac{1}{2\gamma'_e} - \frac{1}{2} \tag{8}$$

Substituting $n' = n'_E$ into Equation (4), we obtain:

$$\Delta E = E_K = E_{\text{electronic}} + E'_S\left(n'_E + \frac{1}{2}\right) - E'_S\gamma'_e\left(n'_E + \frac{1}{2}\right)^2 \tag{9}$$

This equation can be used to calculate the electronic excitation energy $E_{\text{electronic}}$, assuming the dissociation energy E_K and the vibrational constants E'_S and γ'_e are known.

For iodine, the following values apply:

$$E'_S = 0.0159 \text{ eV}, \quad E_K = 0.94 \text{ eV}$$

These values allow for the determination of $E_{\text{electronic}}$ in the iodine molecule

1.2 Colour centers

In ionic crystals, especially alkali halides, electrons are highly localized. These materials have a large energy gap, typically between 5 and 12 eV, separating the valence and conduction bands. Because visible light only carries energy in the range of about 1.5 to 3.5 eV, it cannot excite electrons across this gap. Consequently, pure ionic crystals are transparent and act as electrical insulators.

However, processes such as thermal or chemical treatment, or exposure to ultraviolet light or x-rays, can introduce point defects into alkali halides. These defects (such as vacancies, interstitials, or impurity atoms) are incorporated into the crystal lattice and can trap electrons in bound states that are less localized than those in the perfect lattice. These states lie within the forbidden energy gap and often fall within the visible light spectrum for alkali halides. As a result, the crystals develop characteristic colors (eg, $NaCl$ turns red and KBr turns blue) due to these so-called colour centers. There are several types of colour centers, each historically given a distinct name (see fig. 5).

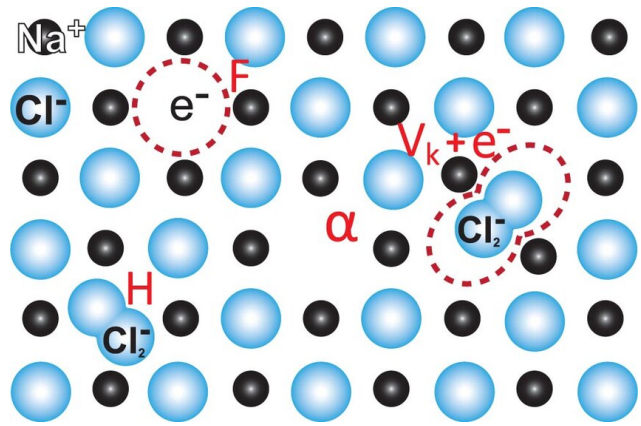


Figure 5: Different crystal defects (colour centers) in alkali halides and their names [1]

1.2.1 F-centers

An F-center (colour center) is a halogen ion vacancy that traps an electron. It is the most common type of defect and will be the focus of further discussion. F-center can be created through X-ray, UV, or electron irradiation, or by annealing in alkali vapor. The process involves excitation of a halogen electron across the band gap, forming an exciton. The halogen atom then shifts, creating a V_K-center: a bound state involving a hole and neighboring halogen ion. Energy released during exciton recombination can generate a Frenkel-pair (H-center), resulting in both an F-center and an H-center. This V_K-mediated pathway is more energetically favorable than direct formation.

Classical Description

The classical description of an F-center assumes that the negative ion vacancy leads to a positive charge on the surrounding positive ions. This charge of e^+ is assigned homogeneously to the surface of a sphere of radius

$$R^3 = \frac{3\sqrt{2}}{16\pi} a^3 \tag{10}$$

where a is the distance between two neighbouring negative ions in the lattice (i.e., the lattice constant of the crystal).[2]

The electric field felt by the electron e^- trapped in the F-center is that of a spherically symmetric charge distribution, thus only depending on the radius r from the center of the sphere:

$$E(r) = \frac{e^+}{4\pi\epsilon_r\epsilon_0} \frac{r}{R^3} \tag{11}$$

where ϵ_0 the permittivity of free space and ϵ_r the relative permittivity of the material. Thus, the force felt by the trapped electron is

$$F(r) = e^- E(r) = \frac{e^- e^+}{4\pi\epsilon_r\epsilon_0} \frac{r}{R^3} = -\frac{e^2}{4\pi\epsilon_r\epsilon_0} \frac{r}{R^3} \tag{12}$$

This is a restoring force linear in r and together with Newton's Second Law $F = ma$, this results in a harmonic motion for the electron with the angular frequency

$$\omega^2 = \frac{e^2}{4\pi\epsilon_r\epsilon_0 m R^3} \tag{13}$$

Using the usual relation $E = \hbar\omega$ for the energy of oscillation and substituting the upper value for ω and using equation (10), we get for the excitation energy

$$E = \frac{\hbar e}{\sqrt{4\pi\epsilon_r\epsilon_0 m R^{3/2}}} = \frac{he}{\pi\sqrt{3\sqrt{2}\epsilon_r\epsilon_0 m a^3}} \tag{14}$$

where we can see that $E \propto a^{-3/2}$.

This excitation energy of the oscillator corresponds to the transition energy E_m that determines the maximum of the absorption line. The classical model of the F-center is thus a harmonic oscillation of the trapped electron in the electric potential of a homogeneously positively charged sphere.

Quantum Mechanical Description

In the simplest quantum mechanical description, we consider the electron to be trapped in a potential box with side lengths of the lattice parameter a:

$$V(x,y,z) = \begin{cases} 0 & \text{for } 0 \le x \le a,\ 0 \le y \le b,\ 0 \le z \le c, \\ \infty & \text{else} \end{cases} \tag{15}$$

By solving the 3-dimensional Schrödinger equation

$$\left[-\frac{\hbar^2}{2m}\Delta + V(x,y,z)\right]\Psi = E\Psi. \tag{16}$$

and using the boundary conditions $\Psi(x,y,z) = 0$ outside the box, we obtain for the wavefunction the simple equation

$$-\frac{\hbar^2}{2m}\Delta\Psi = E\Psi$$

For a cubic box with side lengths a the energy eigenvalues

$$E_{n_1,n_2,n_3} = E_0(n_1^2 + n_2^2 + n_3^2) \tag{17}$$

with

$$E_0 = \frac{\pi^2\hbar^2}{2ma^2} \tag{18}$$

For transitions from the 2p-like first excited state $E_{1,1,2} = 6E_0$ to the 1s-like ground state $E_{1,1,1} = 3E_0$ we obtain the transition energy

$$E_m = 3E_0 = \frac{3h^2}{8ma^2} \propto a^{-2} \tag{19}$$

which is in good agreement with the experimental Mollow-Ivey relation of alkaline halides:

$$E_m \propto a^{-1.84} \tag{20}$$

A more advanced quantum mechanical model assumes a constant, finite potential within the unit cell, and outside it considers the Coulomb potential from nearby cation point charges, along with an additional contribution from the electron's interaction with lattice polarization due to atomic displacements (polaron coupling).[2]

1.2.2 Theory of the line width

Although discrete energy levels suggest narrow absorption bands, the observed bands for F centers are relatively broad due to the electron's interaction with lattice vibrations. This effect is modeled using a configuration coordinate R, representing the displacement of surrounding cations from the color center. In excited states, the electron's wavefunction spreads out more, and R is smallest for the $1s$ ground state. Energy shifts as a function of R form parabolic curves, and together with the Franck-Condon principle (which explains rapid, vertical electronic transitions), this results in broad absorption and emission bands. The farther the parabolas are shifted with respect to each other, the broader these bands are.

Because of the interaction with the lattice having discrete phonon energies $h\nu_0$, one would expect a series of equally spaced absorption lines with spacing $h\nu_0$. However, low-energy phonons from the surrounding lattice also contribute, since the F-center disrupts the lattice periodicity and relaxes selection rules. As a result, the sharp vibrational levels become broadened, supporting the use of the parabolic model for describing the interaction.

1.3 Concentration of F-centers Centers

Analysis of F-center absorption spectra shows that their concentration N is proportional to the integrated absorption. Assuming a moderate band width and a Lorentzian shape, the absorption can be described using the classical Drude model for an oscillator. This gives the integral absorption:

$$\int \alpha \, d\nu = \frac{e^2 N}{4c^2 \epsilon_0 n m_e} \left(\frac{n^2+2}{3} \right)^2 \tag{21}$$

where n is the refractive index and $m + e$ is the mass of the electron.

To determine the absorption coefficient α, the transmission of plane-parallel crystals is typically measured as a function of wavelength. Transmission T is defined as the ratio of the transmitted intensity I_t to the incident intensity I_0 as

$$T = \frac{I_t}{I_0} = \frac{(1-R)e^{-\alpha d}}{1 - R^2 e^{-2\alpha d}}$$

with the reflectivity R defined as

$$R = \frac{(n-1)^2}{(n+1)^2}$$

where n is the refractive index of crystal and d is its thickness.

In case where $R \ll 1$ (which is the case for NaCl and KBr), the denominator in the equation for transmission approaches 1, the Lambert-Beer law holds:

$$I = I_0 e^{-\alpha d} \tag{22}$$

Measuring the transmission with and without colour centers for a crystal of a given thickness d, allows us the calculate the absorption coefficient α, so that we can calculate the concentration N of F-centers using equation 21.

1.4 Experimental Setup

We use a dual-beam spectrometer to obtain transmittance spectrum for all our samples.

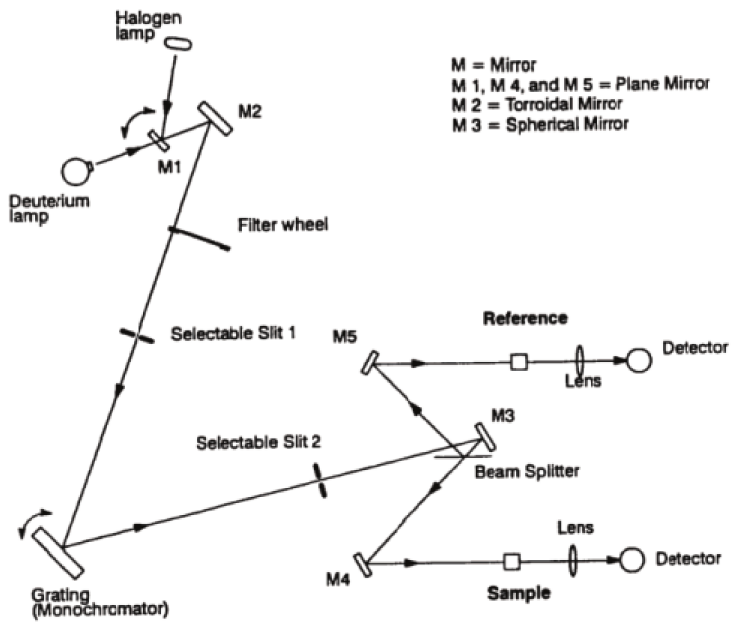


Figure 6: Beam path diagram [2]

The spectrometer uses two light sources: a deuterium lamp for the UV range and a halogen lamp for the visible and near-IR range (above 326 nm). The instrument automatically switches lamps around 400 nm to ensure continuous spectral coverage from 190 to 1100 nm.

A filter wheel eliminates unwanted wavelengths to prevent second-order diffraction from the grating. The light reflects from a diffraction grating. The desired wavelength for measurement is selected by adjusting the angle of the grating.

This selected wavelength passes through a slit, which has an adjustable width. The slit width determines the range of wavelengths that pass through for each measurement, affecting both spectral resolution and light intensity.

The beam is then split into two parts. One beam serves as the reference and does not pass through the sample, while the other beam passes through the sample. The spectrometer detects the intensity of both beams and calculates the difference in intensity at each wavelength, and so we get the transmittance spectrum.

2 Analysis and Results

2.1 Task 1

2.1.1 Scan Speed Variation

To investigate the effect of different scan speeds, we selected a constant slit width of 0.5 nm and recorded measurements of a Holmium oxide glass filter at three different scan speeds.

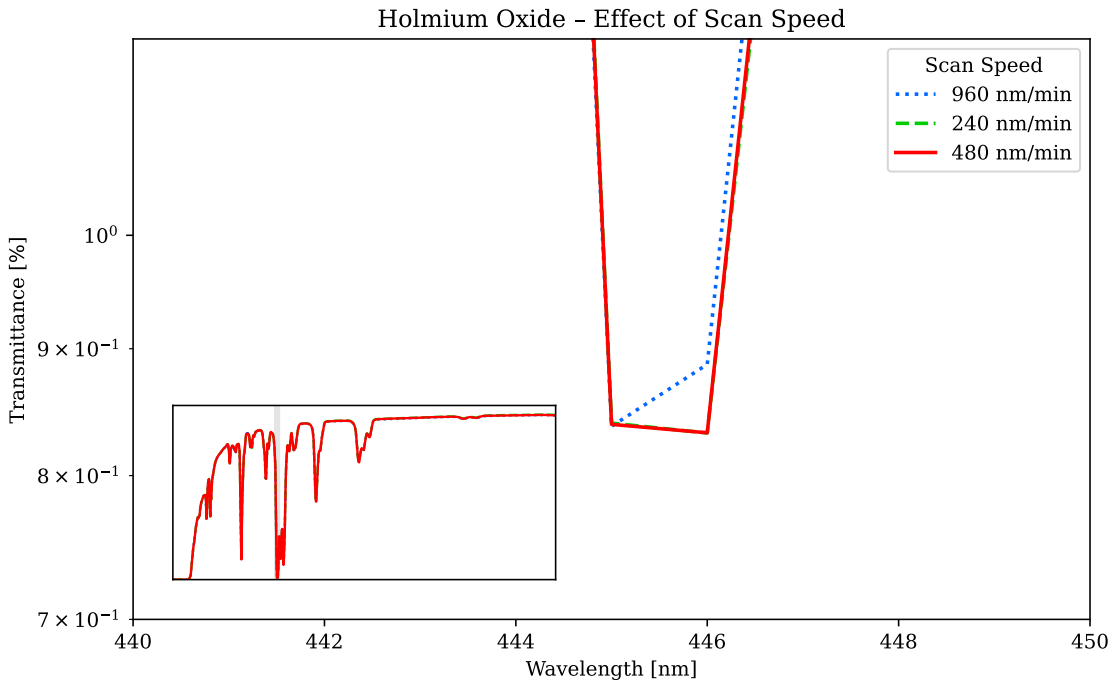


Figure 7: 3 different scan speeds overlayed, zoomed in at a peak with logarithmic scale on y-axis.

There is almost no difference in the scans. A logarithmic scale has been applied to the y-axis, and we zoomed into a peak to notice any differences between measurements. For all scan speeds, the same number of data points was collected—900 points for the range from 200 nm to 1100 nm. Therefore, it makes sense that there is not much difference between the graphs. The only difference is the time taken for each measurement. It can be seen from the graph that 240 nm/min and 480 nm/min have slightly more defined peaks than 960 nm/min. We observed almost no difference between 240 nm/min and 480 nm/min, so we decided to use 480 nm/min for the rest of our measurements.

2.1.2 Slit Width Variation

With a constant scan speed of 480nm/min we did scans of with 3 different slit widths.

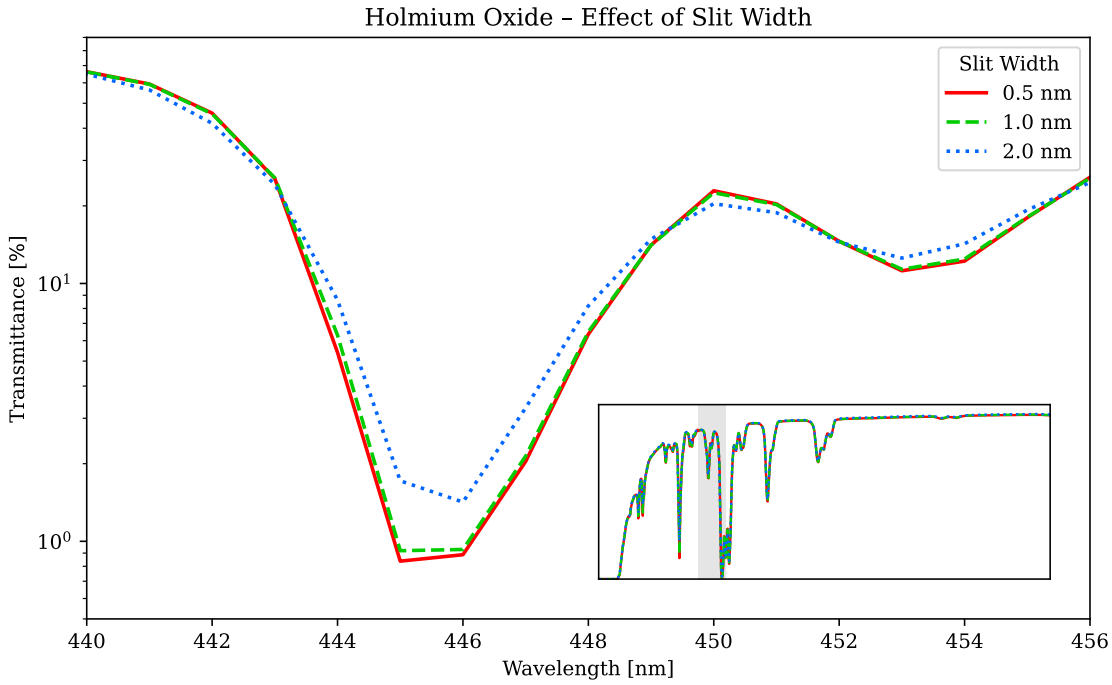


Figure 8: Varying slit width, zoomed in with logarithmic scale on y-axis.

As expected, a smaller slit width provides better resolution for both peaks and dips in the spectrum. This is because a smaller slit width restricts the range of light allowed through for each measurement, resulting in more precise wavelength discrimination. Given the enhanced resolution and the fact that it did not significantly affect the measurement time, we opted to use a slit width of 0.5 nm for the remainder of our measurements.

2.1.3 Wavelength Accuracy

Holmium Oxide is widely used as a reference in spectrometry because it has well-known and sharp absorption bands. We use the Holmium Oxide spectrum to determine the wavelength accuracy of our machine. We found certified absorbing band values from the National Institute of Standards and Technology [3]

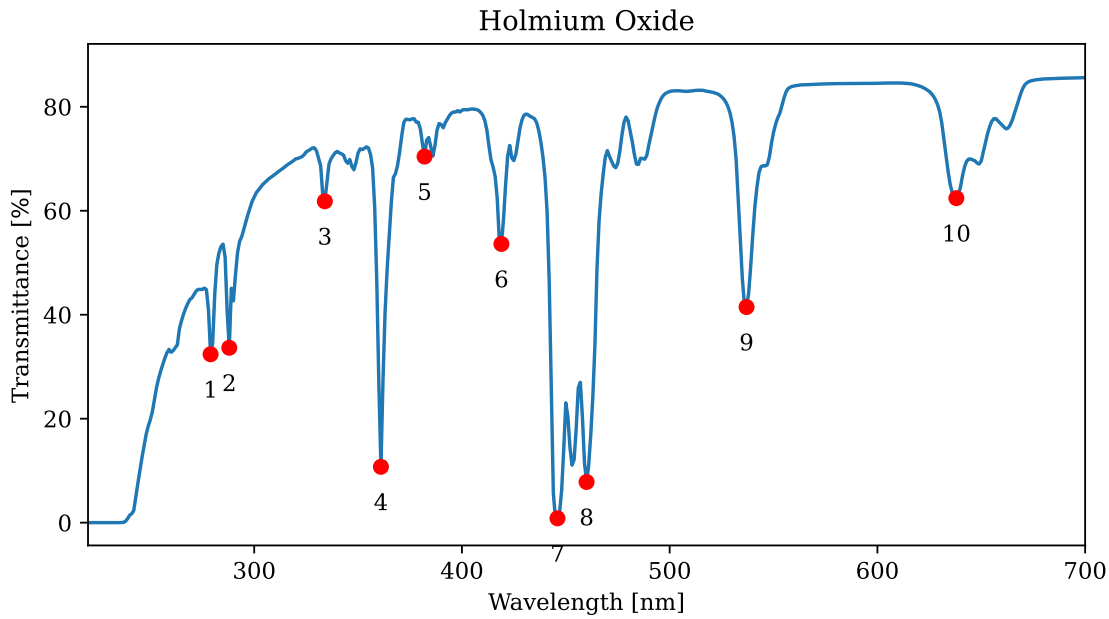


Figure 9: Transmittance spectrum of Holmium Oxide, highlighting labeled absorption bands. The figure is zoomed in to the spectral region containing the transmission minima.

Bands	**Certified Wavelength (nm)**	**Measured Wavelength (nm)**	**Percent Error (%)**
1	279.3	279	0.107
2	287.6	288	0.139
3	333.8	334	0.060
4	360.8	361	0.055
5	385.8	382	0.985
6	418.5	419	0.119
7	453.4	446	1.632
8	459.9	460	0.022
9	536.4	537	0.112
10	637.5	638	0.078
		Average Percent Error:	**0.331**

Table 1: Comparison of Certified Wavelengths and Measured Wavelengths of Holmium Bands in transmittance spectrum.

We get an average percentage error of 0.331% which is acceptable for our measurements.

2.2 Task 2

2.2.1 Transmission Spectra of Glass and Quartz Cuvettes Filled with Both Air and Water

We measured the transmission spectra of glass and quartz cuvettes filled with both air and water. The slit width and scan speed used were those selected in Task 1, and the measurements were performed in the range of 200–1100 nm.

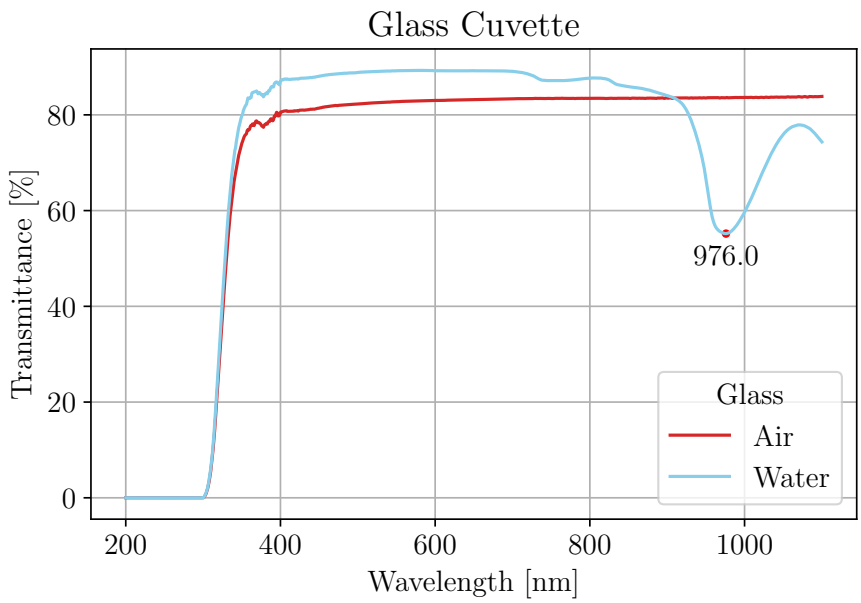


Figure 10: Transmittance Spectra of both Air and Water Filled Quartz Cuvette

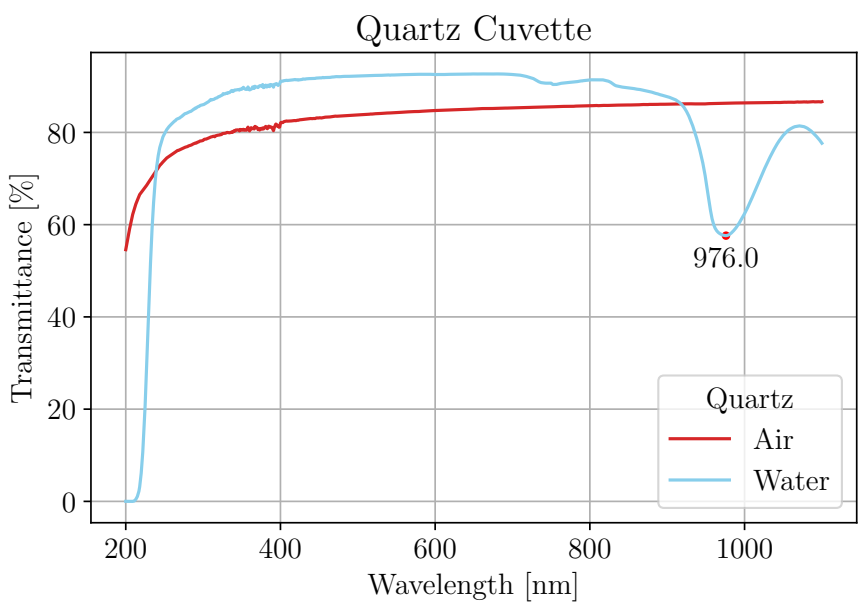


Figure 11: Transmittance Spectra of both Air and Water Filled Quartz Cuvette

In both cases, cuvettes filled with water has higher transmittance. This is because the transmittance is given by:

$$T = 1 - \left(\frac{n_1 - n_2}{n_1 + n_2}\right)^2 \tag{23}$$

The difference in refractive index between glass ($n \approx 1.5$) or quartz ($n \approx 1.46$) and water ($n \approx 1.33$) is much smaller than the difference between these materials and air ($n \approx 1.00$).

Glass blocks a larger portion of the ultraviolet (UV) spectrum, transmitting no light below approximately 300 nm. In contrast, quartz transmits UV light down to around 210 nm.

Both the glass and quartz cuvettes filled with water exhibit a dip at 976 nm, which is likely due to a combination vibration involving the O–H stretching and bending modes in the water molecules, commonly observed in the near-infrared region.

Moreover, we observe that the quartz cuvette filled with air exhibits consistent transmittance throughout the spectrum, with only a slight dip below 210 nm, which could make it suitable for applications such as optical lenses.

2.2.2 Transmission Spectrum of Interference Filter with Varying Angle of Incidence

The transmission spectrum of an interference filter was recorded at four different incident angles, using the same measurement settings and spectral range. Since the transmission is zero between 630 nm and 930 nm, the spectrum was split into two separate graphs for better visualization.

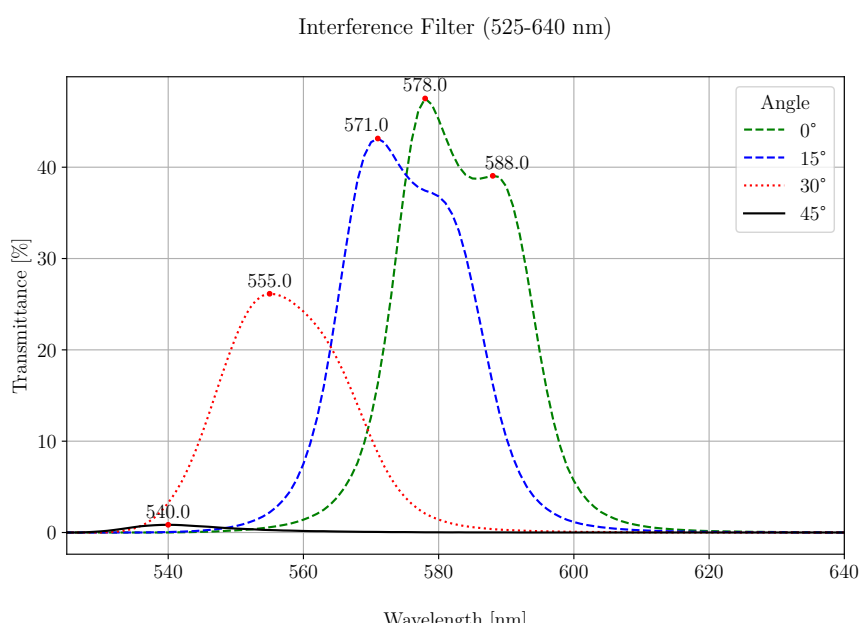


Figure 12: Transmittance Spectrum of an Interference Filter with Varying Angles in the range of (525–640 nm)

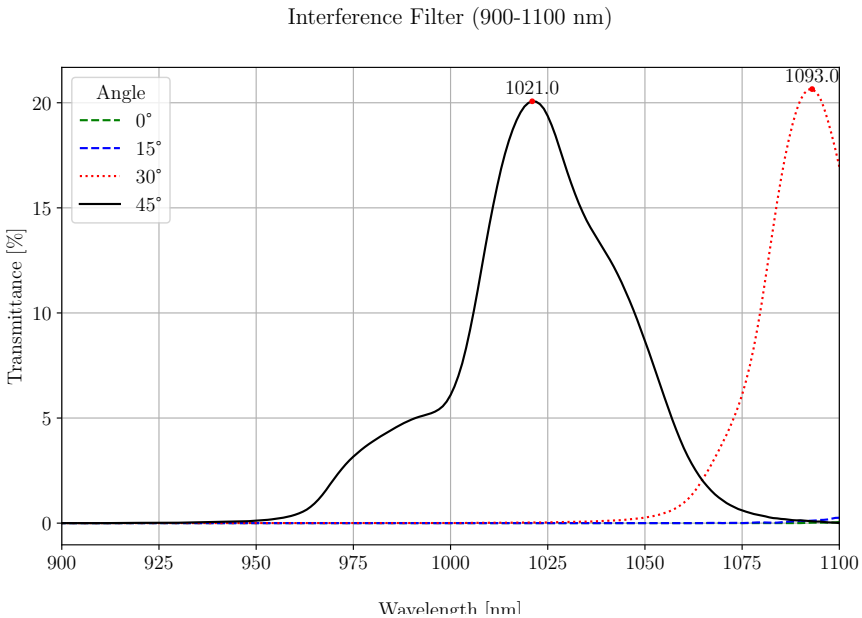


Figure 13: Transmittance Spectrum of an Interference Filter at Various Angles (900–1100 nm)

An interference filter consists of multiple thin layers that allow narrow bands of light to pass through, depending on the angle of incidence. At specific angles, different wavelengths of light pass through the filter with the same optical path length, resulting in constructive interference giving us the peak. From our graphs, we observe that at 30 degrees, the second peak at 1093 nm is almost exactly double the first peak at 555 nm. This is expected, as the second peak is likely due to second-order interference. A similar pattern is observed at 45 degrees. The second-order peaks for the remaining angles are likely outside the measurement range.

2.2.3 Colour Filters

We chose the red, green, and blue colour filters shown below and obtained their transmission spectrum.

Figure 14: Colour Filters Used

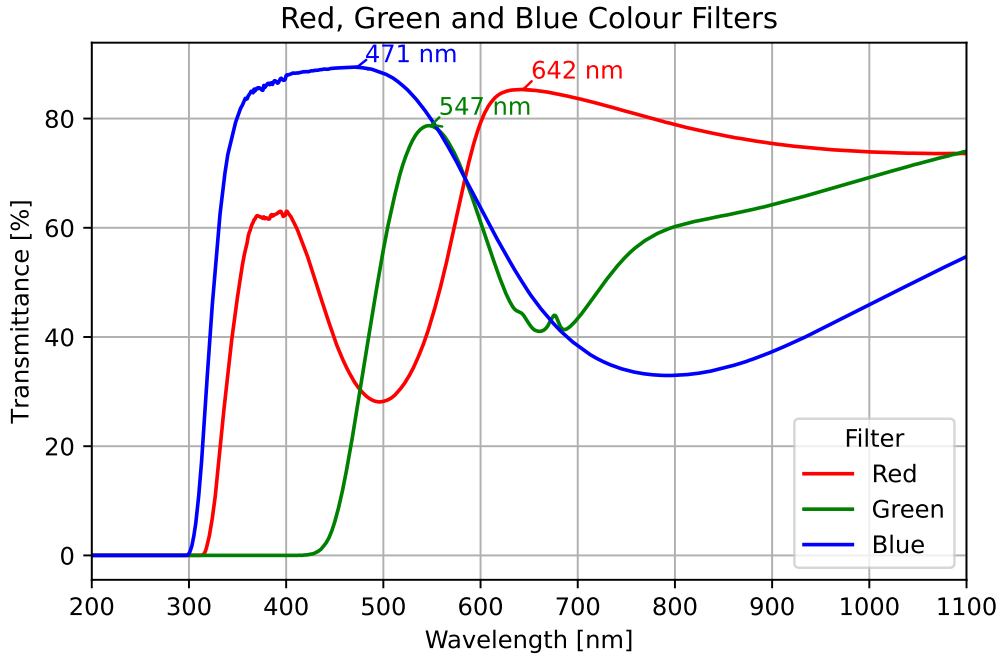


Figure 15: Transmittance Spectrum of 3 Different Colour Filters

The Blue filter has a broad peak centered at 471 nm, which makes sense since the blue range is between 450–485 nm [5]. The broadness of the peak explains why we visually see the filter as pale blue, since a wide range of light is let through.

The peak of the Green filter is also within the expected range of 500–565 nm [5]. It shows an increase in transmission around 800 nm, but this is likely outside the visible range for humans, so we don't see its contribution.

The Red filter has two peaks, with the more significant one at 642 nm, which is within the expected range for red (625–750 nm) [5]. The peak around 400 nm explains the paleness of the filter when seen visually. Since more of the spectrum is let through, the color does not appear saturated but instead gives a light red as can be seen in Figure 14.

The Blue filter starts transmitting light at 300 nm. Since blue/violet is the shortest wavelength visible to humans, we can infer from this spectrum that the visible range for the human eye starts at around 300 nm. And since red is the longest visible wavelength, we suggest that the upper end of the visible range is slightly beyond the peak of the Red filter, around 700 nm.

We can also see that the transmission of these three filters covers almost all of the visible range, which makes sense, as Red, Green, and Blue are primary colors that can be combined to produce every other color in the spectrum through the additive process.

2.3 Task 3

In this task, we aim to determine the convergence energy E_k, the dissociation energy of the ground state E''_D, the dissociation energy of the excited state E'_D, and the electronic transition energy $E_{\text{electronic}}$.

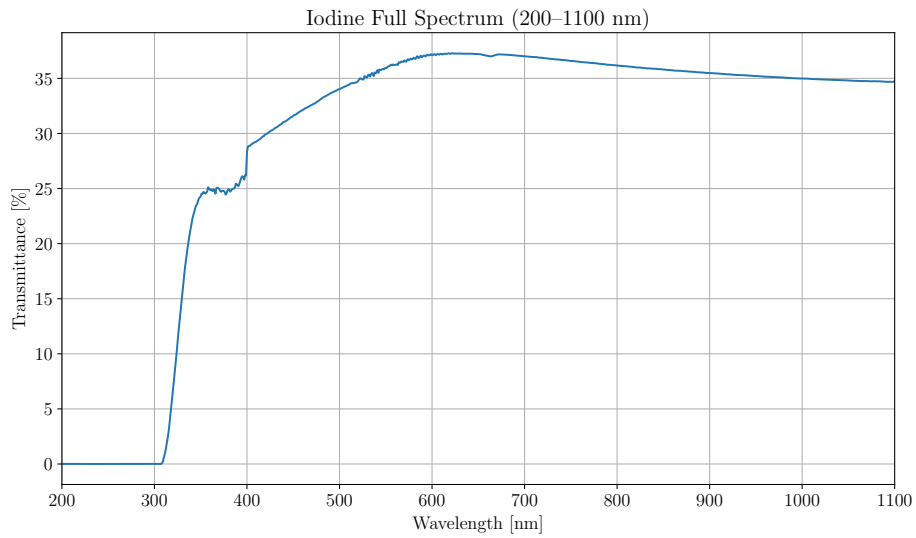


Figure 16: Iodine Vapour Full Transmission Spectrum

We focus on the region with absorptions due to combined effect of electronic and vibrational transitions, and mark the distinguishable dips in the transmission spectrum.

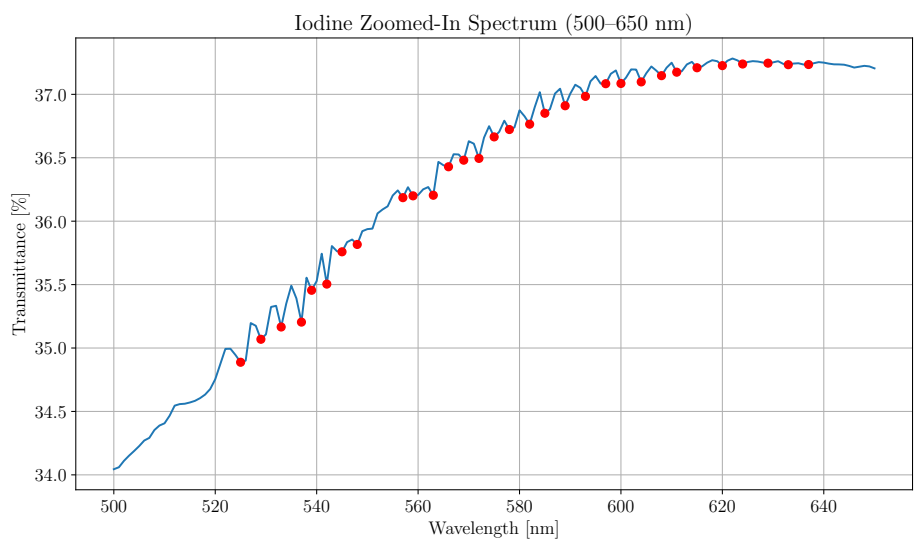


Figure 17: Zoomed-in transmission spectrum of iodine vapour with absorption dips marked.

Then converted the wavelength of the dips to corresponding photon energies using (3).

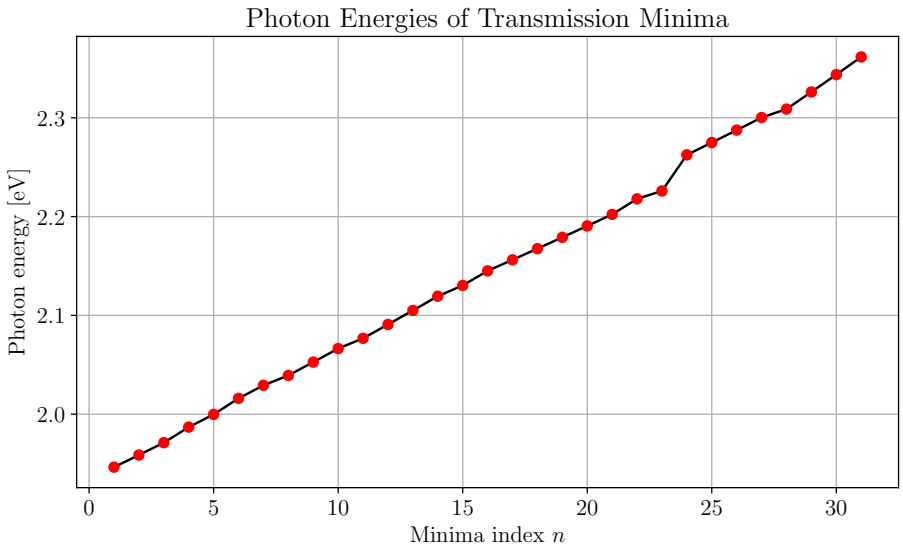


Figure 18: Photon energies corresponding to the transmission minima

It can clearly be seen that there are two distinct linear regions. The first linear region corresponds to transitions from the vibrational ground state ($n'' = 0$), which is the most populated state at room temperature. The shorter, flatt region corresponds to transitions from higher vibrational levels ($n'' > 0$) in the ground electronic state to the excited state, which are less probable. Only the absorption minima originating from the ground vibrational state $n'' = 0$ are plotted, with each point labeled according to the excited vibrational quantum number n'.

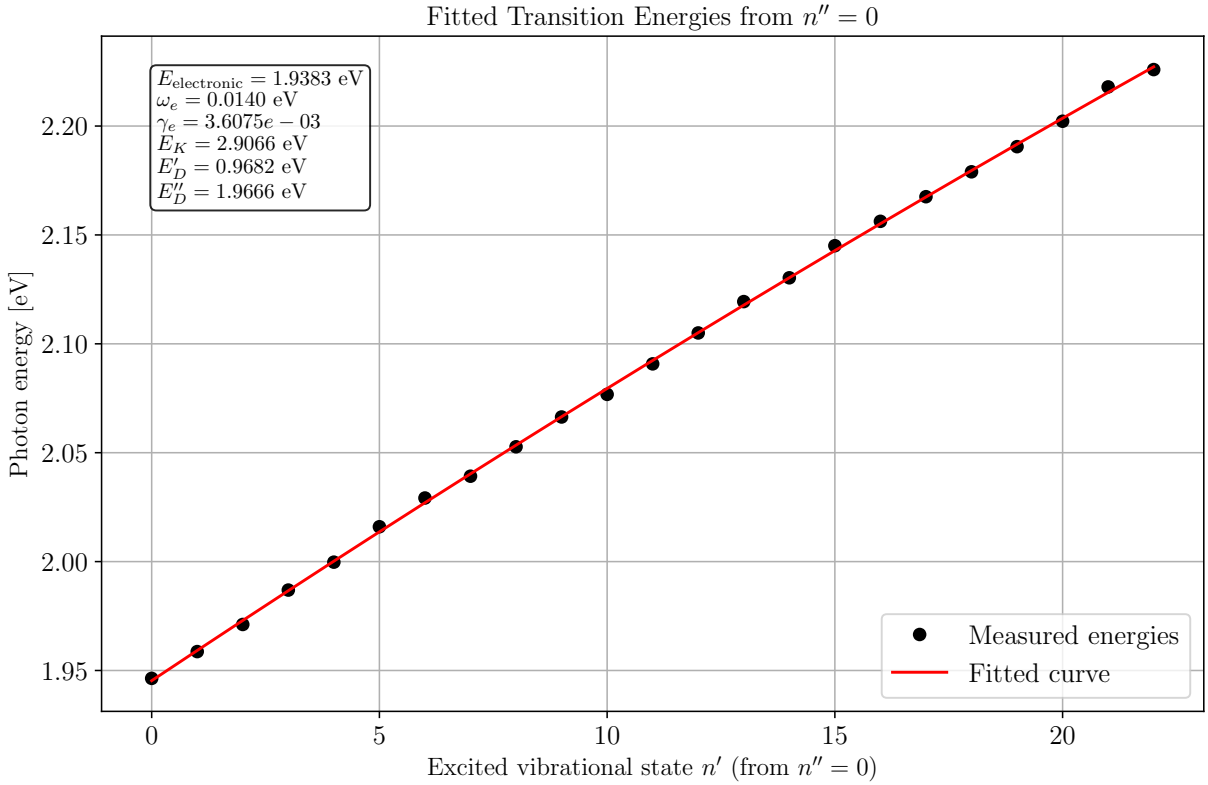


Figure 19: Photon energies for transitions from $n'' = 0$ to excited vibrational states n', fitted using the anharmonic oscillator model

To determine the convergence energy E_K, the energies of the first few transitions from the ground vibrational state $n'' = 0$ to various excited states n' were fitted using the vibrational energy expression in Equation (4). The resulting fit parameters included the vibrational spacing E'_S, the anharmonicity constant γ'_e, and the electronic excitation energy $E_{\text{electronic}}$. Using these, the dissociation energy E_K of the excited state was calculated from Equation (9) by evaluating the energy at the effective vibrational quantum number at dissociation, n'_E, as given in Equation (8).

The dissociation energy E'_D of the excited state was then found as the difference $E_K - E_{\text{electronic}}$, and the dissociation energy of the ground state E''_D was determined by subtracting the known binding energy of the ground state (0.94 eV) from E_K.

2.4 Task 4

For this task, we irradiated a KBr crystal with UV rays for an extended period of time to generate F-centers. Then we measured the transmission spectrum of the crystal four different times, in 10-minute intervals. We used this to calculate the excitation energy and the line width.

In the absence of high energy radiation, the concentration of colour-centers decreases with

time, as the excited electrons return to their ground state. This results in a changing transmittance profile over time, i.e. in a lessening dip in the absorbance region of the color centers (approximately in the region from 450 nm to 850 nm), as seen below in Fig. 20.

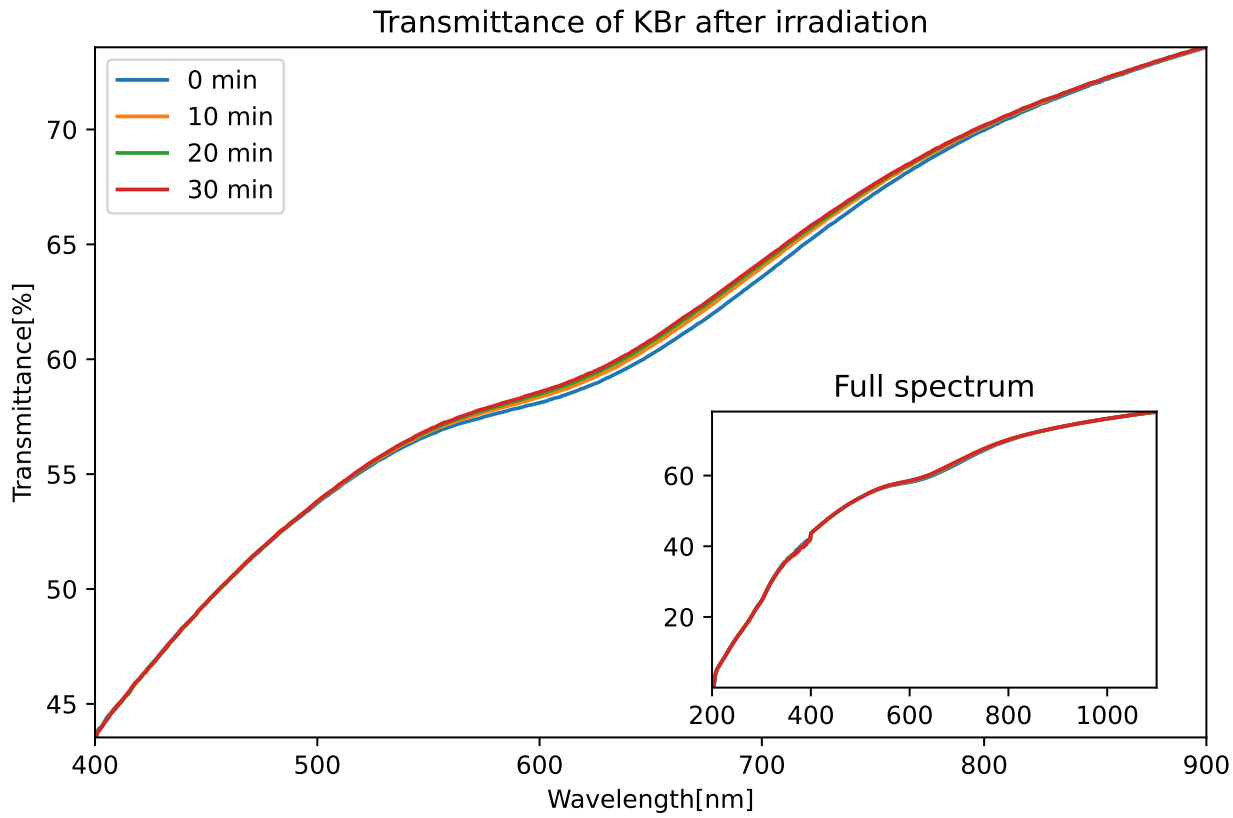
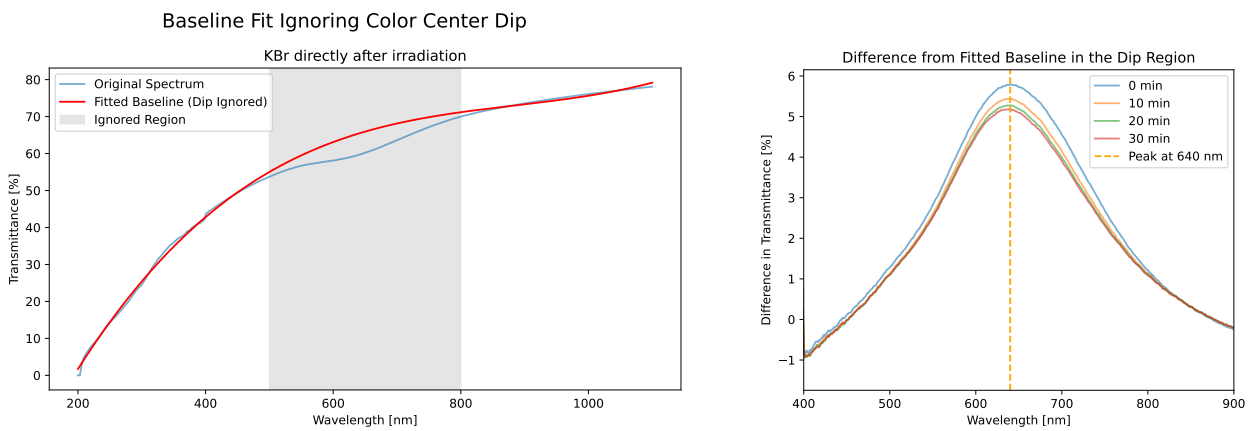


Figure 20: Transmittance Spectrum of KBr at different times, measured in 10-minute intervals

To evaluate the change in the transmittance spectrum compared to a sample with no colour-centers, we fitted a 3-degree polynomial to the spectrum, ignoring the dip region, such that we obtained a reference spectrum. For each measurement we performed a different fit, to account for any changes that arise due to different errors in different measurements. The performed fit can be seen in Fig. 21a for one of the measurements, the others were performed analogously. Then we took the difference in transmittance to this baseline, as seen in Fig. 21b.

All the dips show a peak at 640 nm, from which the excitation energy can be calculated using eq. 3. We obtain the value of 1.94 eV or 3.10×10^{-19} J



(a) (b)

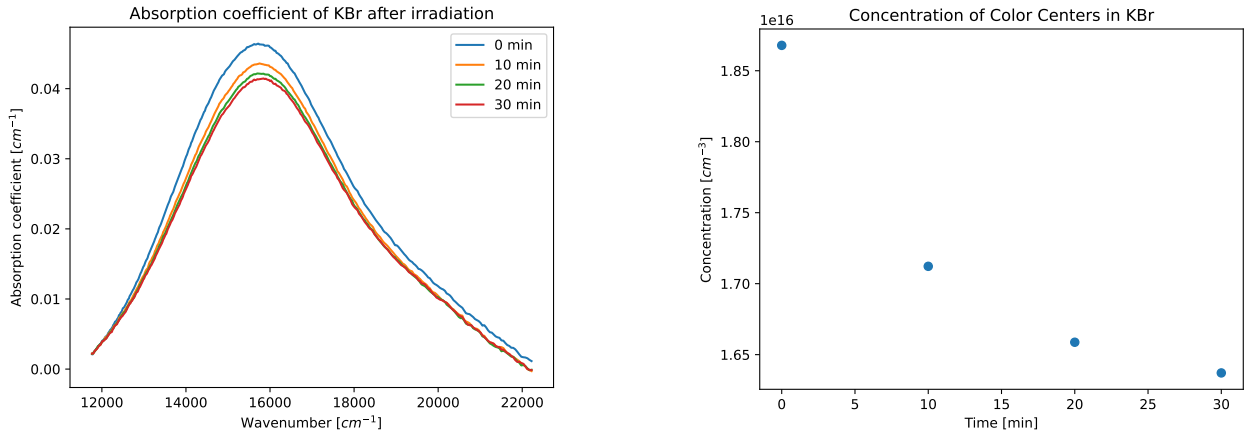
Figure 21: (a) Fitted baseline (polynomial of degree 3) for the measurement directly after irradiation (b) Difference in transmittance to the baseline. All of them have a peak at 640 nm

Using the fitted baseline spectra in wavenumber space, we calculated the absorption coefficient α (see Fig. 22a). Then, using Eq. 21 and using $d = 2\,\text{cm}$ for the thickness of the crystal, we obtained the concentration of colour-centers as they changed over time. The values can be seen in the Table 2 below, together with the Full Width at Half Maximum values of the peaks of difference in transmittance over wavelength (as in Fig. 21b).

Time after irradiation [min]	**FWHM [nm]**	**F-center concentration [cm^{-3}]**
0	207.02	1.87×10^{16}
10	203.16	1.71×10^{16}
20	203.72	1.66×10^{16}
30	204.37	1.64×10^{16}

Table 2: FWHM and F-center concentration as a function of time after irradiation.

As expected, the concentration is decreasing in time, but an exponential fit could not be performed with confidence. This could be improved by waiting longer and taking more measurements, as the concentration stayed quite high even 30 minutes after irradiation.



(a) (b)

Figure 22: (a) The obtained absorption coefficient α over wavenumber, used for the integral in Eq. 21 (b) The obtained concentration of F-centers decaying over time

2.5 Task 5

For this task we were given the data for the transmission spectrum of a NaCl sample before and after UV-radiation (i.e. without and with color-centers). The spectrum can be seen below in Fig. 23.

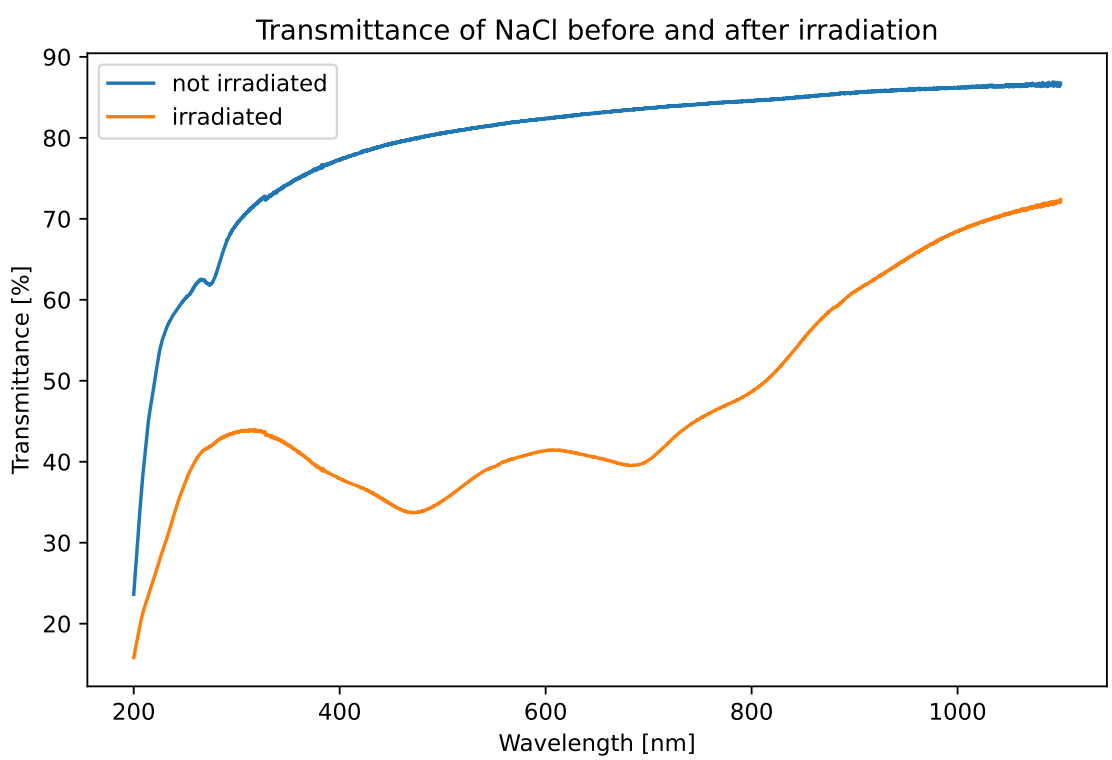
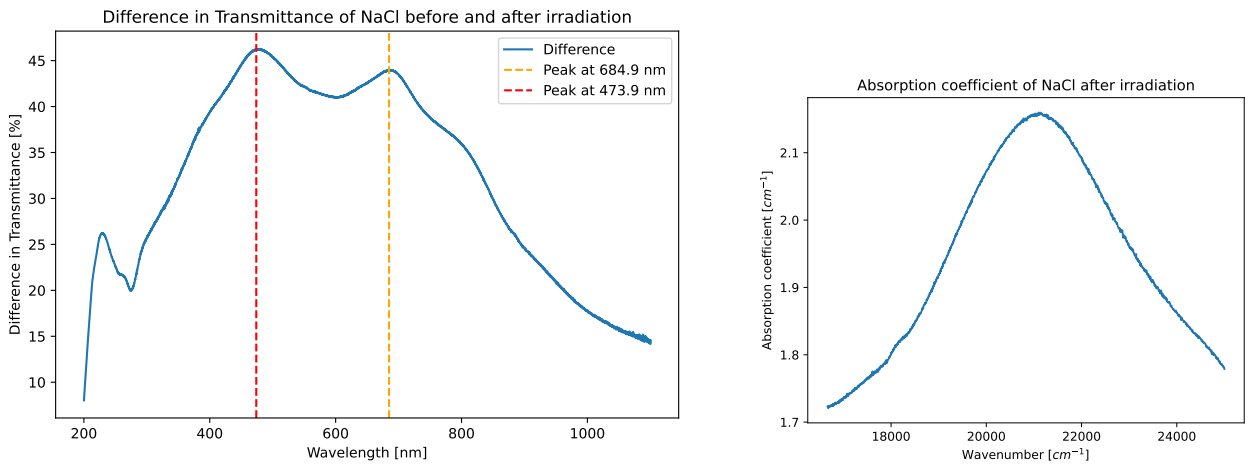


Figure 23: Transmittance Spectrum of NaCl before and after irradiation

This time the effect of the color centers is not that obvious, as we can see a significant drop in transmittance across the whole range of measured wavelengths. The difference in transmittance compared to the reference spectrum can be seen below in Fig. 24a.



(a) (b)

Figure 24: (a) Difference in transmittance of irradiated NaCl to the reference spectrum (b) The obtained absorption coefficient α over wavenumber for NaCl, restricted to the range of 400-600 nm corresponding to the most intense peak

We can see several peaks in the difference in transmittance, the two most prominent peaks being located at 473.9 nm and 684.9 nm, from which the excitation energies can be calculated using eq. 3. The peaks with their corresponding FWHM values and excitation energies can be seen the the table below. The very high FWHM values are the result of the two peaks overlapping.

Peak location [nm]	**FWHM [nm]**	**Excitation energy [eV]**
473.9	5108.12	2.62
684.9	630.43	1.81

Table 3: Peak location, FWHM, and excitation energy for the two most prominent peaks

As simple F-centers are the most common type of colour-center, we can assume that the most prominent peak can be attributed to the presence of these. From this, with the same procedure as in Task 4, we can calculate the concentration of colour-centers. Using $d = 0.4\,\text{cm}$ for the thickness of the crystal (as provided in the manual), we obtain the value of $N = 1.38 \times 10^{18} cm^{-3}$.

The other peaks likely correspond to other types of color centers (eg. pairs or aggregates of F-centers).

2.6 Task 6

For this task, we calculated the excitation energies from KBr and NaCl within the classical and the quantum mechanical models, as in equations 14 and 19:

$$E_{classical} = \frac{he}{\pi\sqrt{3\sqrt{2}\epsilon_r\epsilon_0 m a^3}}$$

$$E_{QM} = \frac{3h^2}{8ma^2}$$

Using the lattice constants $a_{KBr} = 6.69\,\text{Å}$ and $a_{NaCl} = 5.63\,\text{Å}$ from the manual[2] and the values of $\epsilon_{KBr} = 4.88$ and $\epsilon_{NaCl} = 5.90$ for the relative permittivities[4], we obtained the following values:

	$\mathbf{E}_{classical}$ **[eV]**	$\mathbf{E}_{QM}$ **[eV]**	$\mathbf{E}_{experimental}$ **[eV]**
KBr	0.96	2.60	1.94
NaCl	1.11	3.56	2.62

Table 4: Obtained excitation energies for KBr and NaCl with the classical and quantum mechanical approaches compared to the experimental values from Task 3 and 4

In both cases the classical value is lower than the experimental value, while the quantum mechanical value is higher, but gives a closer estimate. The result could be improved by considering a more advanced quantum mechanical model, as discussed in Section 1.2.1.

Using the excitation energies obtained with the quantum mechanical approach, we can also estimate the lattice constants using the same equation 19 to express it as

$$a = \left(\frac{3h^2}{8mE_{QM}}\right)^{1/2}$$

The results are reasonably close to the reference values (see Table 5 below)

	$\mathbf{a}_{reference}$ **[Å]**	$\mathbf{a}_{estimate}$ **[Å]**	**Percentage error [%]**
KBr	6.59	7.63	15.80
NaCl	5.63	6.57	16.63

Table 5: Calculated estimates for the lattice constants of KBr and NaCl compared to the reference values

3 Conclusion

Task 1: Scan speed had little effect on quality, so 480 nm/min was chosen for efficiency. A 0.5 nm slit width offered the best resolution. Wavelength calibration showed strong agreement with certified values (avg. error 0.331%).

Task 2: It could be seen that transmission was higher for cuvettes filled with water because it has a refractive index closer to that of glass and quartz. The quartz cuvette filled with air had a very flat transmission spectrum. We observed the first and some second order angle-dependent constructive interference peaks for the interference filter. The colour filters produced peaks at the expected wavelengths, consistent with their visual appearance.

Task 3: Iodine absorption spectra were analyzed to extract E_K, $E_{\text{electronic}}$, and dissociation energies using anharmonic oscillator fits. Results aligned well with theory, confirming vibrational structure models.

Task 4: UV-irradiated KBr sample showed F-center formation with measurable dips in the transmittance spectrum. Excitation energy, color-center concentration, and line width were calculated. The concentration of color-centers decreased over time, though longer measurements are needed for proper analysis of the decay.

Task 5: Irradiated NaCl showed broad transmittance loss with two key absorption peaks. From these, excitation energies and a color-center concentration of $1.38 \times 10^{18}\,\text{cm}^{-3}$ were determined. Peaks suggest both simple F-centers and more complex defects.

Task 6: Classical and quantum mechanical models were used to estimate F-center excitation energies. The quantum model better matched experiment and could be used to calculate lattice constants within 16% of reference values, though refinements could improve accuracy.

4 References

References

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